Integration & Applications

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int x^4\,dx$.

$\frac{x^5}{5} + C$.

2 **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int \sqrt{x}\,dx$.

$\int x^{1/2}\,dx = \frac{2}{3}x^{3/2} + C$.

3 **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int \frac{1}{x^3}\,dx$.

$\int x^{-3}\,dx = -\frac{1}{2}x^{-2} + C = -\frac{1}{2x^2} + C$.

4 **[Foundational]** • **Target: Antiderivatives of polynomials**

Find $\int (4x^3 - 6x + 5)\,dx$.

$x^4 - 3x^2 + 5x + C$.

5 **[Foundational]** • **Target: Antiderivatives of polynomials**

Find $\int (2x + 7)\,dx$.

$x^2 + 7x + C$.

6 **[Foundational]** • **Target: Antiderivatives of polynomials**

Find $\int (x^2 - 4x + 3)\,dx$.

$\frac{x^3}{3} - 2x^2 + 3x + C$.

7 **[Foundational]** • **Target: Antiderivatives of trigonometric and exponential functions**

Find $\int \cos x\,dx$.

$\sin x + C$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int \sin x \, dx$.

$-\cos x + C$.

9 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int e^x \, dx$.

$e^x + C$.

10 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_0^2 3x^2 \, dx$.

$[x^3]_0^2 = 8$.

11 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_1^3 2x \, dx$.

$[x^2]_1^3 = 9 - 1 = 8$.

12 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_{-1}^2 3x^2 \, dx$.

$[x^3]_{-1}^2 = 8 - (-1) = 9$.

13 [Intermediate] • Target: Definite integrals with polynomial integrands

Evaluate $\int_0^2 (x^2 + 1) \, dx$.

$\left[\frac{x^3}{3} + x\right]_0^2 = \frac{8}{3} + 2 = \frac{14}{3}$.

14 [Intermediate] • Target: Definite integrals with polynomial integrands

Evaluate $\int_1^3 (2x^2 - 3x + 1) \, dx$.

$F(x) = \frac{2x^3}{3} - \frac{3x^2}{2} + x$. $F(3) = 18 - 13.5 + 3 = 7.5$. $F(1) = 0.667 - 1.5 + 1 = 0.167$. Integral $= 7.5 - 0.167 = \frac{22}{3}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Definite integrals with polynomial integrands • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Evaluate $\int_{-2}^1 (x^2 - x) \, dx$.

$F(x) = \frac{x^3}{3} - \frac{x^2}{2}$. $F(1) = -\frac{1}{6}$. $F(-2) = -\frac{8}{3} - 2 = -\frac{14}{3}$. Integral $= -\frac{1}{6} + \frac{14}{3} = \frac{9}{2}$.

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16 [Challenging] • Target: Definite integrals of trigonometric and exponential functions • Cross-Lesson: 6.01 Limits & Derivatives

Evaluate $\int_0^{\pi/2} \cos x \, dx$.

$[\sin x]_0^{\pi/2} = 1 - 0 = 1$.

17 [Challenging] • Target: Definite integrals of trigonometric and exponential functions • Cross-Lesson: 8.01 Exponentials & Logarithms

Evaluate $\int_0^{\pi} \sin x \, dx$.

$[-\cos x]_0^{\pi} = -\cos(\pi) - (-\cos(0)) = 1 - (-1) = 2$.

18 [Challenging] • Target: Definite integrals of trigonometric and exponential functions • Cross-Lesson: 9.01 Statistics & Probability

Evaluate $\int_0^1 e^x \, dx$.

$[e^x]_0^1 = e - 1 \approx 1.72$.

19 [Challenging] • Target: Area under a curve • Cross-Lesson: 1.01 Algebra & Equations

Find the area under $y = x^2$ from $x = 0$ to $x = 3$, using a definite integral.

Area $= \int_0^3 x^2 \, dx = \left[\frac{x^3}{3}\right]_0^3 = \frac{27}{3} = 9$.

20 [Challenging] • Target: Area under a curve • Cross-Lesson: 2.01 Functions & Transformations

Find the area under $y = 4 - x^2$ from $x = -2$ to $x = 2$.

$F(x) = 4x - \frac{x^3}{3}$. $F(2) = 8 - \frac{8}{3} = \frac{16}{3}$. $F(-2) = -8 + \frac{8}{3} = -\frac{16}{3}$. Area $= \frac{16}{3} - \left(-\frac{16}{3}\right) = \frac{32}{3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Area under a curve

The graph shows $y = x^2$. Find the area under the curve from $x = 0$ to $x = 3$, using a definite integral.

a. 9 ✓

b. 18

c. 3

d. 27

$\left[\frac{x^3}{3}\right]_0^3 = 9$.

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22 [Foundational] • Target: Area under a curve

The graph shows $y = 4 - x^2$. Find the area under the curve from $x = -2$ to $x = 2$.

a. $\frac{16}{3}$

b. 32

c. 16

d. $\frac{32}{3}$ ✓

$\left[4x - \frac{x^3}{3}\right]_{-2}^{2} = \frac{16}{3} - \left(-\frac{16}{3}\right) = \frac{32}{3}$.

23 [Foundational] • Target: Word problems: integration in context

The graph shows a particle's velocity $v(t) = 8t - 2t^2$ for $0 \le t \le 4$. Find the particle's displacement (the area under the curve).

a. $\frac{64}{3}$ ✓

b. 16

c. 32

d. 64

$\int_0^4 (8t - 2t^2)\,dt = \frac{64}{3}$ m.

24 [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int \sqrt[3]{x}\,dx$.

a. $\frac{3}{2}x^{2/3} + C$

b. $\frac{4}{3}x^{4/3} + C$

c. $\frac{3}{4}x^{4/3} + C$ ✓

d. $\frac{1}{3}x^{-2/3} + C$

$\int x^{1/3}dx = \frac{x^{4/3}}{4/3} + C = \frac{3}{4}x^{4/3} + C$.

25 [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int 7\,dx$.

a. C

b. $7x + C$ ✓

c. $7 + C$

d. $\frac{7x^2}{2} + C$

The antiderivative of a constant k is $kx + C$.

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26 [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int \frac{1}{\sqrt{x}}\,dx$.

a. $2\sqrt{x}+C$ ✓

b. $\frac{1}{2}\sqrt{x}+C$

c. $-2\sqrt{x}+C$

d. $\sqrt{x}+C$

$\int x^{-1/2}dx = \frac{x^{1/2}}{1/2}+C = 2\sqrt{x}+C.$

27 [Foundational] • Target: Antiderivatives of polynomials

Find $\int (3x^3-2x)\,dx$.

a. $3x^4-x^2+C$

b. $\frac{3x^4}{4}-x^2+C$ ✓

c. $\frac{3x^4}{4}-2x^2+C$

d. $\frac{x^4}{4}-x^2+C$

Integrate each term.

28 [Foundational] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int 2\cos x\,dx$.

a. $2\sin x+C$ ✓

b. $-2\sin x+C$

c. $2\cos x+C$

d. $\sin(2x)+C$

$\int \cos x\,dx = \sin x.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int 3\sin x\,dx$.

a. $3\sin x+C$

b. $-3\sin x+C$

c. $3\cos x+C$

d. $-3\cos x+C$ ✓

$\int \sin x\,dx = -\cos x.$

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30 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int (e^x + 2)\,dx$.

a. $e^x + 2 + C$

b. $e^x + 2x + C$ ✓

c. $xe^x + 2x + C$

d. $e^x + C$

$\int e^x\,dx = e^x$; $\int 2\,dx = 2x$.

31 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int (\sin x + \cos x)\,dx$.

a. $-\cos x - \sin x + C$

b. $\cos x + \sin x + C$

c. $\cos x - \sin x + C$

d. $-\cos x + \sin x + C$ ✓

$\int \sin x\,dx = -\cos x$; $\int \cos x\,dx = \sin x$.

32 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_2^5 3\,dx$.

a. 3

b. 9 ✓

c. 6

d. 15

$3 \times (5 - 2) = 9$.

33 [Intermediate] • Target: Definite integrals of trigonometric and exponential functions

Evaluate $\int_0^{\pi/2} \sin x\,dx$.

a. $\frac{\pi}{2}$

b. -1

c. 0

d. 1 ✓

$[-\cos x]_0^{\pi/2} = -0 + 1 = 1$.

34 [Intermediate] • Target: Definite integrals of trigonometric and exponential functions

Evaluate $\int_0^{\pi} \cos x\,dx$.

a. π

b. -2

c. 0 ✓

d. 2

$[\sin x]_0^{\pi} = 0 - 0 = 0$.

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35 [Intermediate] • Target: Signed area and regions below the axis

Evaluate $\int_0^4 (x-4)\,dx$ and state whether the region is above or below the x-axis.

a. 8; above

b. -8; above

c. 8; below

d. -8; **below** ✓

$\left[\frac{x^2}{2} - 4x\right]_0^4 = 8 - 16 = -8$; negative means below the x-axis.

36 [Intermediate] • Target: Properties of definite integrals

Given $\int_1^5 f(x)\,dx = 20$ and $\int_1^5 3\,dx = 12$, find $\int_1^5 [f(x) - 3]\,dx$.

a. -8

b. 8 ✓

c. 32

d. 17

$20 - 12 = 8$.

37 [Intermediate] • Target: Properties of definite integrals

Given $\int_0^4 f(x)\,dx = 9$ and $\int_0^4 g(x)\,dx = 5$, find $\int_0^4 [f(x) + g(x)]\,dx$.

a. $\frac{9}{5}$

b. 45

c. 4

d. 14 ✓

$9 + 5 = 14$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Finding a function from its derivative • Cross-Lesson: 9.01 Statistics & Probability

A function satisfies $f'(x) = 2x + 7$ and $f(-1) = 4$. Find $f(x)$.

a. $x^2 + 7x + 10$ ✓

b. $2x^2 + 7x + 10$

c. $x^2 + 7x - 10$

d. $x^2 + 7x + 4$

$f(x) = x^2 + 7x + C$; $f(-1) = 1 - 7 + C = 4 \Rightarrow C = 10$.

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39 [Challenging] • Target: Average value of a function • Cross-Lesson: 1.01 Algebra & Equations

Find the average value of $f(x) = x^2 - 1$ on $[1, 3]$.

a. $\frac{8}{3}$

b. $\frac{10}{3}$ ✓

c. $\frac{20}{3}$

d. 6

$\frac{1}{2}\left[\frac{x^3}{3} - x\right]_1^3 = \frac{1}{2}\left(6 - \left(-\frac{2}{3}\right)\right) = \frac{10}{3}$.

40 [Challenging] • Target: Word problems: integration in context • Cross-Lesson: 2.01 Functions & Transformations

A particle moves along a line with velocity $v(t) = 8t - 2t^2$ (m/s) for $0 \le t \le 4$. Find the particle's displacement over this interval.

a. 64

b. $\frac{64}{3}$ ✓

c. 32

d. $\frac{192}{3}$

$\int_0^4 (8t - 2t^2)\,dt = \left[4t^2 - \frac{2t^3}{3}\right]_0^4 = \frac{64}{3}$ m.

41 [Challenging] • Target: Word problems: integration in context • Cross-Lesson: 3.01 Sequences & Series

Water flows into a tank at a rate of $r(t) = 30 - 3t$ liters per minute, for $0 \le t \le 10$. Find the total volume of water that flows in during this time.

a. 300

b. 150 ✓

c. 450

d. 100

$\int_0^{10} (30 - 3t)\,dt = [30t - 1.5t^2]_0^{10} = 150$ liters.

42 [Challenging] • Target: Word problems: integration in context • Cross-Lesson: 4.01 Trigonometry

A factory's production rate is $p(t) = 50 + 4t$ items per hour, for $0 \le t \le 8$. Find the total number of items produced during this 8-hour period.

a. 528 ✓

b. 656

c. 450

d. 400

$\int_0^8 (50 + 4t)\,dt = [50t + 2t^2]_0^8 = 528$ items.

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43 [Challenging] • Target: Integration & Applications • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find $\int -x^4\,dx$.

a. $-\frac{1}{5}x^5 + C$ ✓

b. $-\frac{1}{4}x^4 + C$

c. $-x^5 + C$

d. $-\frac{1}{6}x^5 + C$

$-\frac{1}{5}x^5 + C$.

44 [Challenging] • Target: Integration & Applications • Cross-Lesson: 6.01 Limits & Derivatives

Find $\int -3x^4\,dx$.

a. $-3x^5 + C$

b. $-\frac{3}{5}x^5 + C$ ✓

c. $-\frac{1}{2}x^5 + C$

d. $-\frac{3}{4}x^4 + C$

$-\frac{3}{5}x^5 + C$.

45 [Challenging] • Target: Integration & Applications • Cross-Lesson: 8.01 Exponentials & Logarithms

Find $\int (-2x + 5)\,dx$.

a. $-2x + 5 + C$

b. $-x^2 + 7x + C$

c. $-x^2 + 5x + C$ ✓

d. $-x^2 - 5x + C$

$-x^2 + 5x + C$.

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